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Element A: Presentation and Justification of the Problem

It has been estimated that pushing constitutes nearly a quarter of manual material handling efforts in industrial settings (Baril-Gingras and Lortie 1995). This scope is expanded in the context of caster carts for pushing, which are used in various fields to carry loads.

However, cart pushing presents issues in user experience.

It is estimated that treatment of lower back pain (LBP) and shoulder injury costs over $50 billion and $7 billion annually in direct costs, respectively. As employers recognize the risk of lifting tasks, the prevalence of pushing and pulling tasks in manual material handling (MMH) has increased (Weston *et al*. post-2016).

Despite this tradeoff, there is a significant correlation between occupational pushing and musculoskeletal disorders. Overextension of the musculoskeletal system while pushing a cart is correlated with shoulder, neck, and lower back pain. LBP is one of the most common musculoskeletal disorders and occupational incapacity can lead to economic loss (Giagloglou *et al*. 2015). Occupation incapacity can secondarily impact businesses and supply chains reliant on these employees.

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Frymoyer *et al*. (1980) correlated the prevalence of multiple occupational risk factors and incidence of LBP. The pushing risk factor was present in 20% of men and 31% of women with LBP, compared to 2% of men and women without LBP.

These musculoskeletal disorders are observed across many domains. In nursing, 34% of women expressed concern for musculoskeletal disorders from pushing beds or heavy trolleys (Estryn-Behar *et al*. 1990). Firefighting, construction work, floor mopping, and garden raking likewise have been found to incorporate intensive use of pushing and pulling (Hoozemans 1998). Indeed, overexertion injuries account for 35% of firefighter injuries, many related to pushing, at an average cost of $9,715 per claim (IAFC 2018). These issues spill over to the consumer in the form of shopping carts. Recent research shows ergonomics to be a significant factor in user satisfaction with pushing carts (Estember and Berdan 2017). A 2020 study reports that 39% of injuries in the motor vehicle part manufacturing industry were related to push/pull movements (Yang et al., 2020).

As such, solutions that reduce musculoskeletal stress are of particular interest to a broad group of stakeholders ranging from industrial workers to the insurance companies representing them (i.e., secondary stakeholders). Solutions to optimize cart ergonomics may be of use to business owners aiming to optimize customer satisfaction or employers aiming to minimize workplace accidents. Case studies in truck manufacturing plants highlighted a

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need to frequently consult individuals at all levels of the supply chain, factory management steering group, engineers, technicians, and operators (Zare *et al*. 2020).

Low handle heights (e.g. at the hip level) present musculoskeletal concerns. Research supports that low handles are associated with greater compressive forces on the lower back, presenting a risk for musculoskeletal disorders (Lee *et al*. 1991).

Beyond lower back compression, existing literature suggests that handle elevation is inversely related to the minimum coefficient of friction between the user's feet and the ground surface (Lee *et al*. 1992).

Lee's study further indicates that low handles present slip risks on tile, concrete, and steel surfaces.

Indeed, research suggests that although waist-level handles are optimal in minimizing spinal compression forces on the lower back in pulling, shoulder-level handles are optimal for pushing-related tasks (Argubi-Wollesen *et al.* 2017). An increase in force applied likewise is associated with increased shear force on the lower back. Existing research suggests that the leaning posture, by requiring a greater net applied force, induces stress and shear force on the lumbar spine (Pinupong *et al*. 2020).

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As such, there is clearly a need for solutions to minimize lower back injury while simultaneously reducing slip potential. Lee *et al*. 1991's methodology highlights the need to consider a user's stature when determining an optimal handle height. Thus, there is no inherent "one size fits all" handle height that can minimize these consequences for all users.

The particular problem we aim to address is increased musculoskeletal stress for Mr. Kiser, the group's primary stakeholder. Mr. Kiser (as well as other teachers) often must push carts carrying textbooks, testing materials, or other items across a campus. Typical weight loads can exceed 300 lbs.

Field research indicates that Mr. Kiser utilizes carts with a typical waist-level height (approx. 33in from the ground). Mr. Kiser is (6'3") and suffers from lower back pain (slipped discs). As such, he must bend down and suffer back pain when pushing the waist-level handle. As Pinupong *et al*. (2020) suggests, this leaning posture induces further force on the lower back.

Having to use many different carts carrying different loads, Mr. Kiser wants any solution to relieve this pain to be removable and re-attachable for use on separate carts.

Simultaneously, back pain minimization must be accounted for during reattachment and removal. That is, the act of bending down to implement a solution has the potential to worsen lower back pain.

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